

WASP-163b

R-2172

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_250504 · created 2026-08-13T00:40:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460799.797 +/- 0.061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.043
Transit depth (Rp/Rs)^2	2.19 +/- 1.26 [%]
Orbital Inclination (inc)	89.43 +/- 2.75
Airmass coefficient 1 (a1)	1.661 +/- 0.037
Airmass coefficient 2 (a2)	-0.1518 +/- 0.0047
Scatter in the residuals of the lightcurve fit is	2.14 %
Best Comparison Star	None
Optimal Aperture	4.15
Optimal Annulus	14.65
Transit Duration (day)	0.058 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.043 vs 0.119 (deviation 0.67σ)
Duration fit vs published	0.058 d vs 0.0942 d (deviation 1.1σ)
Mid-time O–C	-11.5 min
Depth SNR	3.4
Residual scatter	2.14 %

Light curve

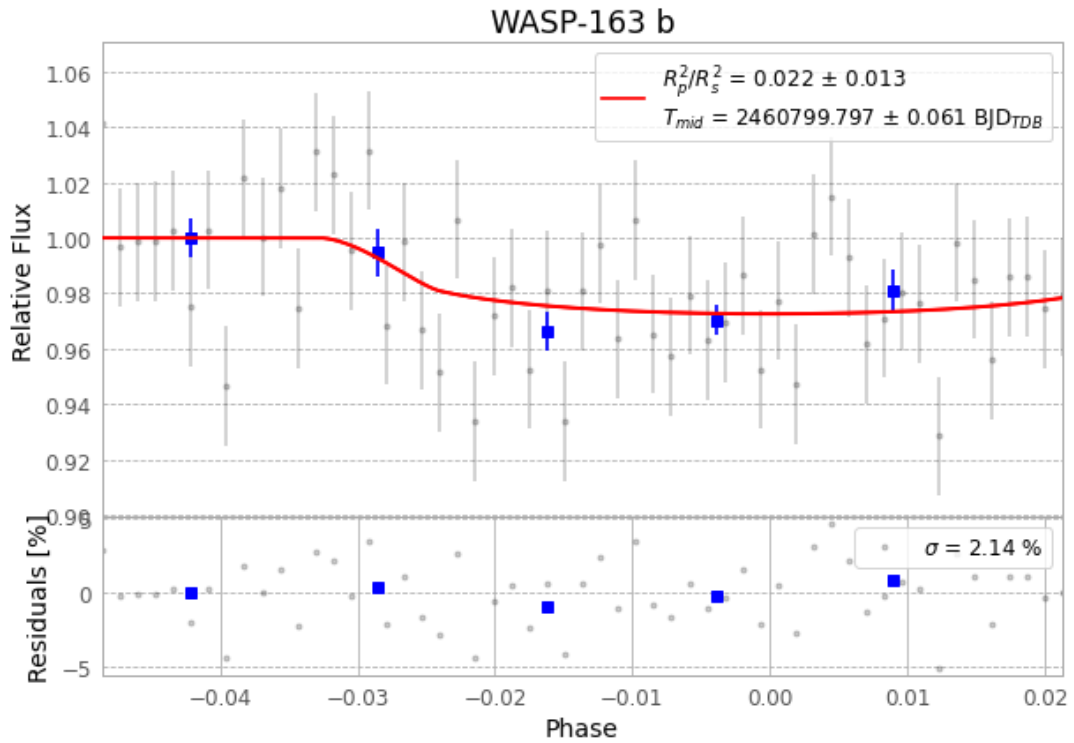


Figure 1 — FinalLightCurve_WASP-163 b_2025-05-03.png (SHA-256 aed314ad606c7bd7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 167], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 516ebec55abc63f1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

55 FITS frames (36 MB), WASP-163250504050615.FITS → WASP-163250504074816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-250504.log (264e1b2b72cf...), FinalLightCurve_WASP-163 b_2025-05-03.png (aed314ad606c...), FinalLightCurve_WASP-163 b_2025-05-03.csv (9144f7f7d178...)

Compute resources

Wall time 92.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:40Z.